

Abhay Raj

AI Engineer | LLM Applications | Retrieval-Augmented Generation | Applied Machine Learning

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Education

Master of Science, Business Intelligence & Analytics – Stevens Institute of Technology (Expected May 2026)

Relevant Coursework: Machine Learning, Web Mining, Data Mining, Optimization & Process Analytics, Multivariate Data Analysis

Technical Skills

Programming: Python, JavaScript, SQL

AI / ML: Transformers, LLM Applications, Retrieval-Augmented Generation (RAG), Prompt Engineering, NLP, Text Classification, Model Evaluation

Frameworks & Libraries: PyTorch, Hugging Face Transformers, Scikit-Learn, Pandas, NumPy

AI Systems: Agentic Workflows, LLM Tooling, Data Pipelines, Experimentation

Tools: Git, Streamlit, REST APIs, Jupyter Notebook

Projects

Exocortex Debugger – AI-Assisted Code Debugging System

Developed an AI-powered debugging platform that executes Python code in a sandbox environment, parses stack traces, and uses LLM reasoning to generate root-cause explanations and corrected code suggestions. Built modular components for execution, traceback extraction, and LLM orchestration with a Streamlit interface.

Rationale Agent – Incident Intelligence System

Built an LLM-driven assistant that converts unstructured engineering incident reports into structured root-cause analysis with ranked hypotheses and supporting evidence extraction. Implemented reasoning workflows and automated claim extraction to improve incident triage.

Predictive Infrastructure Demand Orchestration – Multi-Cloud AI System

Designed an AI-based decision system to forecast compute demand and recommend optimal cloud resource allocation across regions and providers using workload simulation and cost optimization strategies.

BERT-Based Sentiment Analysis Platform

Implemented a transformer-based NLP pipeline using BERT to classify sentiment from social media text data including preprocessing, tokenization, model fine-tuning, and evaluation.